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COMMUNITY ECOLOGY – ORIGINAL RESEARCH

Spatial distribution of damage affects the healing, growth, and morphology of coral

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Abstract

Many predators and herbivores do not kill their prey, but rather remove or damage tissue. Prey are often able to heal or regenerate this lost tissue. If the prey are modular organisms (e.g., some plants and cnidarians), regeneration is frequently influenced by other modules interconnected to damaged ones. For example, many coral predators remove tissue from colonies consisting of many polyps, and these polyps often share resources with their neighbors. Thus, the distribution of tissue loss on a coral colony could affect the coral's response. I hypothesized that spatially aggregated damage might be slow to heal due to competing demands on nearby polyps. To explore the spatial patterns of corallivory and their implications, I conducted: (1) field surveys documenting the spatial distribution of lesions on corals; (2) field experiments testing the effect of the distance between lesions on coral tissue healing, skeletal growth, and morphology; and (3) field surveys relating corallivore presence to coral growth and morphology. In the field surveys, lesions were aggregated at multiple spatial scales, and most lesions had other lesions within 2 cm. When lesions were near one another, coral tissue regeneration was depressed, although there was no effect on whole colony growth. After a year, however, linear extension was lower in the neighborhood of the lesions. Additionally, gastropod corallivores (*Coralliophila violacea*) with low movement decreased coral growth and increased coral topographical complexity. These results suggest that corallivores that create clusters of coral damage have a greater effect on coral growth and recovery from damage than corallivores that spread damage throughout the colony.

Keywords Coral damage · *Coralliophila violacea* · Massive *Porites* · Spatial distributions

Introduction

Many biogenic habitats (e.g., groves of aspen trees, grasslands, kelp forests, and coral reefs) consist of modular or colonial organisms. While each unit (e.g., plant ramet, or coral polyp) is distinct and contributes to the overall growth of the organism (genet or colony), the units often are interconnected. These connections allow the units to communicate with one another and share resources (e.g., Marbà et al. 2002; Alpert et al. 2003; Stuefer et al. 2004). The benefits of these connections often depend on the underlying habitat and resource distributions, and as a result of their interconnectedness, modules in better habitats can benefit neighboring modules in resource-poor or deleterious microhabitats. For example, buried ramets of *Psammochloa villosa* that are connected to unburied ramets survive better and grow larger than ramets without these connections (Yu et al. 2004), and when in spatially heterogeneous water conditions, connected *Alternanthera philoxeroides* ramets with connections grow more than those without (You et al. 2016). Due to connections among ramets, the entire plant benefits from ramets in the high-quality habitats.

Connectivity among units not only helps spread resources to stressful microhabitats, but also benefits the organism when the stressful condition is a damage event. In the case of damage to only part of the organism, units unaffected by damage can contribute resources to facilitate the regrowth and healing of the damaged units. Thus, the health,

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proximity, and connectivity of neighboring units to the damaged unit may influence the rate of recovery of the entire genet. Damaged units with higher connectivity to healthy units should have a greater healing potential than damaged units with few healthy neighbors. For example, shoots of clonal plants with intact connections to other shoots receive nutrients from these other shoots and recover from defoliation better than those with severed connections (Schmid et al. 1988), and connected ramets grow more in the face of extreme temperatures and insect grazing than unconnected ramets (Adonsou et al. 2016). Because of the role of healthy units in healing damaged units of colonial organisms, some researchers actually advocate the use of lesion healing rate as an indicator of the health of an entire colony (e.g., in corals: Fisher et al. 2007). However, the presence of some healthy units is not always enough to ensure recovery, and the sharing of resources could be detrimental. For example, if the damage is too great and/or the condition of the colony too poor, reallocation of resources to the damaged region could harm the colony (Hellström et al. 2006), for example, by reducing competitive ability or reproductive capacity.

The implications of partial damage and ramet connectivity in a modular organism are especially well illustrated in corals, which are colonial, modular organisms composed of interconnected polyps. Coral damage has a wide range of causes such as storms, pathogens and predators (corallivores), and these often result in the partial death of the colony and/or the loss of biomass (Henry and Hart 2005). Lesions can provide substrate for competitors to settle (Titlyanov et al. 2005) or entry points for disease (Katz et al. 2014). Thus, rapid healing of lesions is essential for the continued growth and survival of the colony, and large amounts of healthy tissue can be involved in the recovery process (Oren et al. 1997b).

Corals can respond to damage by rapidly regenerating tissue to cover a lesion from connecting areas of the coral (D'Angelo et al. 2012) through a four-step process (Palmer et al. 2011). Newly regenerated tissue usually extends out from the border of the damaged area [although if a coral has deep, healthy tissue that was undamaged, new tissue might appear to come from the interior of the lesion (Edmunds and Yarid 2017)]. This regrowth of tissue is facilitated by the gastro-vascular canals through which polyps share nutrients, and the availability of resources within the colony has long been known to affect the amount of regeneration (Bak 1983; Meesters et al. 1992). Therefore, coral lesions typically exhibit an exponential rate of recovery as they deplete the pool of available resources (e.g., (Meesters et al. 1994; van Woesik 1998; Denis et al. 2011; Edmunds and Yarid 2017)), and most of the lesion is healed quickly, although complete closure can be a longer-term process. For example, in massive *Porites,* lesions are no longer detectable after 19 months (van Woesik 1998), yet few are completely healed within 6 months (Denis et al. 2011).

Many factors influence the rate of regeneration, likely by influencing the amount of available resources (see review by Henry and Hart 2005). For example, healing is more likely in smaller lesions and lesions with a greater perimeter-to-area ratio (Bak and Steward-Van Es 1980; Meesters et al. 1997; Oren et al. 1997a), presumably because areas heal faster when they have a greater number of neighboring, healthy polyps. The link between available resources and regeneration is further highlighted by the role of symbiont communities and heterotrophy: corals with symbiont communities that provide resources exhibit higher regeneration rates than corals without (DeFilippo et al. 2016; Burmester et al. 2017), and fed corals with symbiont communities show greater healing than those that are unfed (Burmester et al. 2018). Additionally, bleached corals exhibit lower regeneration rates (Meesters and Bak 1993; Mascarelli and Bunkley-Williams 1999), and corals with bumpier morphotypes can exhibit higher regeneration rates than massive or columnar morphologies (Van Veghel and Bak 1994). Finally, abiotic conditions such as temperature can influence regeneration rates due to available resources (Edmunds and Lenihan 2010; Denis et al. 2011, 2013).

While resources from healthy tissue are used in the recovery from damage (Oren et al. 1997b), the extent of involvement differs among corals. For example, in *Favia favus*, the entire colony contributes resources in the regeneration of damaged tissue (Oren et al. 1997b). In other corals, there is a limited distance over which coral polyps can share nutrients. For example, in massive *Porites*, feeding by the gastropod corallivore *Coralliophila violacea* creates carbon sinks to which carbon is translocated, but carbon is only translocated from polyps within 5 cm of the damage (Oren et al. 1998). This limited area of involvement in regeneration could explain why colony size is not necessarily a predictor of regeneration (Bak and Steward-Van Es 1980; Wahle 1983).

On reefs, corals are affected by a diverse community of corallivores, and different species of corallivores consume different parts of the coral with varying degrees of severity (Rotjan and Lewis 2008). Some consume mucus (e.g., *Alpheid* shrimp), tissue (consumed by corallivorous butterflyfishes and gastropods such as *Coralliophila* spp. and *Drupella* spp.), or both the tissue and the skeleton (consumed by triggerfish and parrotfish). Lesions created by these different predators affect coral skeletal growth and tissue regeneration differently; excavating predators create the most damage, while browsing predators that consume only mucus have the least effect (Cameron and Edmunds 2014). Predators also differ in their behavior, leaving lesions of various shapes and do not feed uniformly in space (Rotjan and Lewis 2008; Jayewardene et al. 2009).

Given the documented effects of lesion size, shape, and location on the response of a coral to damage, it is likely that the configuration of lesions affects the coral's response as well. If lesions are clustered, the proximity of neighboring lesions will likely influence the amount of resources available to regenerate coral tissue at a focal lesion, and lesions that are clustered or aggregated near other lesions will likely have lower regeneration rates than isolated lesions. Indeed, damage allotted between multiple lesions rather than one large (or two adjacent lesions) has greater healing (DeFilippo et al. 2016).

Corals with lesions that are dispersed might heal lesions more quickly, but temporarily allot fewer resources to the skeletal growth of the colony, leading to a trade-off of increased tissue regeneration and decreased coral skeletal accretion. By heavily depleting a smaller pool of resources, aggregated damage might have longer-term effects on skeletal growth. Additionally, if areas far from lesions grow normally, this could lead to differential growth within a colony which might alter the morphology of the coral. While many studies document the role of various factors (e.g., lesion size and shape) in regeneration and link these factors to the availability of resources that can be shared throughout the colony, few consider how the spatial configuration of damage influences the response of the coral.

Thus, the spatial configuration of lesions is a largely unexplored, but potentially important factor that can influence the ability of coral to recover from damage. I designed a study to address the following questions using massive *Porites*. (1) How is coral damage distributed on the reef (e.g., are lesions spatially aggregated or overdispersed)? (2) How does lesion proximity affect tissue regeneration and skeletal growth? (3) How does location on the coral colony and distance between lesions affect spatial patterns of coral growth? (4) How does chronic damage affect coral morphology?

Methods

Field surveys of coral damage

I quantified the spatial patterns of coral damage (both size/shape of the lesion and the spatial patterns of the lesions) by conducting photoquadrat surveys in the backreef of the north shore of Moorea, French Polynesia, where massive *Porites* was the dominant coral. At each of two sites on either side of Cook's Bay at depths of 2–3 m, I haphazardly selected two large, focal corals in the backreef lagoon. The backreef of the north shore of Moorea has a strong, unidirectional flow running from the reef crest to the shore (Hench et al. 2008). Therefore, I ran four 25 m transects in the cardinal directions from the bommie to capture variation both parallel to and perpendicular to flow. Along each transect, I randomly selected ten locations where I took a photograph of a 0.25 m by 0.25 m quadrat. Rather than photograph each from top down, the bottom edge of the quadrat was placed against the coral at the random point generated, and the rest of the quadrat positioned against the coral. I excluded photos that were not dominated by live massive *Porites* coral, or that were not in focus, including 95 photoquadrats in the analysis.

In each photograph, a technician identified lesions on massive *Porites* (areas of visible skeleton within continuous coral tissue), and circled each lesion using Image J (Fig. 1, Online Resource). For each lesion, the technician quantified its area and perimeter, and the *x*–*y* coordinate of the lesion centroid (i.e., center). Lesions varied by depth (Cameron and Edmunds 2014), but largely appeared to be due to corallivory rather than breakage. Two technicians analyzed images, but there was no detectable effect of technician identity when comparing scorings by the two, so I combined all data in the final analyses with a total of 3957 lesions.

With the data from these surveys, I quantified: (1) the spatial distribution of lesions among quadrats; (2) the spatial distribution of lesions within quadrats; and (3) the shape (area: perimeter) of the lesions. To determine how lesions were distributed among quadrats, I modeled the counts among all quadrats using generalized linear models relating the number of lesions to the area of coral in the quadrat and including site as a factor to control for differences in corallivore communities or other damaging factors that might differ between the two sites. I fit both Poisson and negative binomial distributions and compared the models using AIC. I also tested for statistical overdispersion using a test developed by Cameron and Trivedi (1990), and examined the distribution of lesions across sites by comparing each of the sites sampled using a Tukey's post hoc test.

To study spatial point patterns of lesions within a quadrat, I used the Clark–Evans nearest neighbor method (Clark and Evans 1954), which compares the observed mean nearest neighbor distance to that expected under complete spatial randomness, and classified each quadrat as either "clustered", "random" or "even". Lesions classified "clustered" would have more lesions in close proximity than lesions in quadrats with "even" lesion distributions. I also examined the distribution of nearest neighbor distances between lesion centroids across all quadrats. To describe the shape of the lesion, I compared the area of the lesion to its perimeter by fitting the equation $P = \pi\sqrt{2}\sqrt{\frac{A}{c\pi} + \frac{A}{\pi}}$, where P is an approximation of the perimeter, A is the area of the lesion, and c is the ratio of the width and height of the ellipse. Values of c near 1 indicate circular lesions (relatively equal heights and widths and a high area:perimeter ratio). Larger values of c indicate a narrower lesion (unequal heights and widths and a lower area: perimeter ratio).

Effect of damage distribution on lesion healing and whole colony growth

I conducted a field experiment to determine how the spacing between tissue lesions affected lesion healing (i.e., tissue regeneration) and skeletal growth of the colony. I collected 40 small massive *Porites* colonies (6–7 cm diameter) without existing damage near the reef crest on the north shore of Moorea, French Polynesia, placed them in individual ziplock bags, transported them to the Richard B. Gump South Pacific Research Station, and kept them in a flow-through seawater table. Each coral was attached to plastic mesh with Z-Spar marine epoxy. After 24 h, I determined the skeletal mass of the colony using buoyant weight (Davies 1989) and randomly assigned each coral to one of four experimental treatments: no damage, or two lesions with centroids separated by either 1.2 cm, 3.5 cm, or 6 cm. I used a Waterpik and 1 cm diameter circular stencil to make lesions on the tops of the corals and removed all tissue from within the lesion area (Fig. 2, Online Resource). I handled all corals similarly regardless of treatment, and I weighed corals after damaging to ensure that their skeletal mass was the same as prior to damage.

I deployed corals in the field by mounting them to cinderblocks arranging them in a line perpendicular to the current. Corals were blocked in space and time, with one replicate of each treatment in each block. After 20 and 39 days, I returned corals to the lab to weigh and photograph them. I analyzed photographs using Image J to document changes in lesion size at each time point. To determine the effects of damage on coral growth, I tested three hypotheses with a priori orthogonal contrasts at each time point: (1) is there a significant effect of damage on coral growth? (undamaged corals vs. corals that received damage); (2) do adjacent lesions respond differently than lesions that are separated? (corals with lesion centers separated by 1.2 cm vs. corals with lesions with centers separated by at least 3.5 cm); (3) does the degree of spacing between non-adjacent lesions influence healing or growth? (corals with lesions centers separated by 3.5 cm vs. corals with lesions separated by 6 cm). I analyzed these contrasts in R using linear mixed effects models using the lme4 (Bates et al. 2015) and lmerTest (Kuznetsova et al. 2017) packages. I treated coral ID and block as random effects, time as a fixed effect, and tested the three hypotheses along with interactions. I removed one coral from the growth analysis because it was dropped during transport, and three corals from the lesion healing analysis due to poor quality of the photographs. For all analyses, I analyzed data in R (version 3.3.2).

Effect of acute damage distribution on coral linear extension

While the first field experiment tests the short-term effect of tissue damage on the skeletal growth of the entire colony, it does not provide long-term data on the colony growth response at different locations within the coral colony (e.g., growth where tissue was removed, or adjacent to where tissue was removed). Therefore, I conducted a second field experiment to test the effect of the distance between tissue lesions on the growth of specific parts of the colony. I collected 60 massive *Porites* corals approximately 5 cm in diameter near the reef crest and returned them to the lab. I damaged the corals with a Waterpik (as in the previous experiment) with lesion centroids 1.2 cm and 3.5 cm apart. Control corals were handled similarly, but not damaged.

To create a reference line from which to measure spatially explicit growth, I placed the corals in a bath of 15 mg/L of Alizarin Red S dye for 2 days. Alizarin Red S is a vital stain that is incorporated into the skeleton as the coral polyps deposit calcium carbonate. Because I damaged the corals before placing them in the dye, the wounded regions did not incorporate large amounts of dye. After removing the corals from the dye, I attached them to a plastic base, buoyantly weighed and photographed each coral, and deployed them in the field as in the first experiment. I placed corals attached to cinderblocks in two sandflats on the backreef of the north shore of Moorea.

After 1 year, I collected the corals, and placed them in a bleach solution to remove the tissue. I then used an angle grinder to cut the corals in half, ensuring that the cuts passed through the lesion locations. I sanded and dampened the corals before photographing them so that growth lines were clear. To analyze the images, I used Image J and marked damaged areas (indicated by a low density of Alizarin), traced the dye line, and traced the outer surface of the skeleton. The region between the dye line and the outer surface represented skeletal accretion during the course of the experiment. To measure growth, I traced > 100 growth lines per colony (each demarcating the edge of a polyp as it accreted $CaCO_3$) from the outer edge of the dye to the edge of the colony (Fig. 3, Online Resource). I modeled linear extension as a function of location on the coral colony (e.g., edge vs. top of the colony). I did this by measuring the angle of growth relative to the vertical growth line. I removed 15 corals due to death prior to the end of the experiment (seven control, ten near, and eight far).

I analyzed the data using the nlme package in R, and included the angle of growth as a fixed effect. I included the individual coral as a random effect, fit several models (e.g., growth as linear, quadratic, vs. cubic functions of angle) (Table 3), and selected the best model using AIC.

Effect of chronic damage on coral morphology

While the first two experiments test the effects of acute damage on the response of the coral, many coral predators inflict chronic damage that might have greater effects on the coral by repeatedly drawing resources from the same areas of healthy of tissue. Therefore, I evaluated the effects of chronic damage on coral morphology by comparing long-term patterns of coral growth in regions near the corallivorous gastropod, *C. violacea,* with coral growth in regions that lacked this predator. Oren et al. referred to this snail as a "prudent sessile feeder", because it feeds in one location for extended periods of time and creates carbon sinks (no snails moved during the 4-month experiment: Oren et al. 1998), similarly a related species, *Coralliophila abbreviata* rarely moves on mounding colonies (Hayes 1990). While it is known that *C. violacea* reduces *Porites* growth rate (Clements and Hay 2018; Hamman 2018), and related snails in the Atlantic are associated with decline in coral tissue (Miller 2001; Baums et al. 2003) and recovery from bleaching (Shaver et al. 2018), no studies have examined the effects of grazing on coral growth within a coral colony.

I found 12 massive *Porites* corals on the north shore of Moorea, French Polynesia approximately 1 m in diameter with *C. violacea* present. I randomly selected one snail on each colony, and found a similar location on the same colony with respect to height and flow that was at least 20 cm from the nearest snail. At each location, I quantified coral morphology, using a triangular piece of plexiglass (30 cm per side) with a 20 cm by 20 cm grid in the center. The grid contained 20 pre-drilled holes at random locations. The plate was centered at the snail (or control location) and laid flat upon the coral (resting on at least three points). I then measured the perpendicular distance from the coral to the plate at each of the 20 locations and used the variance of these distances to quantify surface rugosity (i.e., "bumpiness"); corals with high variances were more topographically complex than corals with low variance. I remeasured each coral after 1 year (using a nail as a reference point) and calculated the change in variance.

I also measured linear extension of each coral near and far from the snail, by inserting a nail at each location and measuring the amount of vertical growth between the initial and final surveys. While corals created new tissue and extended up the nail, I only measured the vertical extension based upon the surface of the adjacent colony.

I tested for a difference in both linear extension and the change in variance between corals with and without snails using a mixed effects model in lme4 (Bates et al. 2015) and lmerTest (Kuznetsova et al. 2017). I treated coral ID as a random effect, and snail presence as a fixed effect.

Results

Surveys of coral damage

Coral lesions were variable at multiple spatial scales. Across all spatial scales (both between the two sites on either side of Cook's Bay and within each site) there was significant overdispersion ($Z = 1.92$, $P = 0.027$), indicating that lesions were spatially clustered. Additionally, lesion density varied among sites. The two sites east of Cook's Bay had significantly fewer lesions than one of the sites to the west of Cook's Bay, but were similar to the second western site.

Coral lesions were also abundant. The number of lesions per quadrat ranged from 2 to 611, with a mean of 42 lesions per quadrat, or from 0.01 lesions/cm^2 to 0.96 lesions/cm^2 with a mean of 0.12 lesions/cm^2 (Fig. 1a). Between quadrats, lesions were very clustered, with some corals having extremely high densities of lesions, although several had lower levels (Fig. 1a). The model with the best fit for the number of lesions was the negative binomial fit including site (Table 1).

Within quadrats, lesions showed a variety of spatial distributions. Lesions on 12 quadrats indicated spatial clustering, 74 were indistinguishable from spatial randomness, and 9 had a demonstrably even distribution of lesions. Even though the spatial point pattern of lesions was variable, lesions were sufficiently dense to be in close proximity to other lesions. The centers of 75% of lesions were < 1.76 cm from the center of the nearest neighboring lesion (Fig. 1b). Lesions were approximately elliptical with the major axis exceeding the minor one by 15% (Fig. 1c: $c = 1.15 \pm 0.01$; estimate ± SE), and this shape appeared to be independent of overall lesion size, although the size of the lesions were also variable (Fig. 1d).

Effect of damage distribution on lesion healing and colony growth

Lesions further apart showed increased rates of tissue regeneration (Fig. 2a). At the midpoint of the experiment, tissue regeneration was 81% higher on corals with lesion centers separated by at least 3.5 cm compared to lesions with centers 1.2 cm apart ($t_{23.1} = -2.8$, $P = 0.011$), however, there was no difference between lesions with lesions separated by 3.5 cm and 6 cm ($t_{22.2} = 0.26$, $P = 0.8$). However, these effects of lesion spacing, which were present at 20 days, were not detectable by the conclusion of the experiment (day 39 $t_{23.8} = 1.9$, $P = 0.068$).

Skeletal growth showed a slightly different pattern than tissue regeneration (Fig. 2b). Lesions did not significantly decrease skeletal growth relative to undamaged corals at the midpoint ($t_{31.2} = 1.8$, $P = 0.089$) or conclusion of the

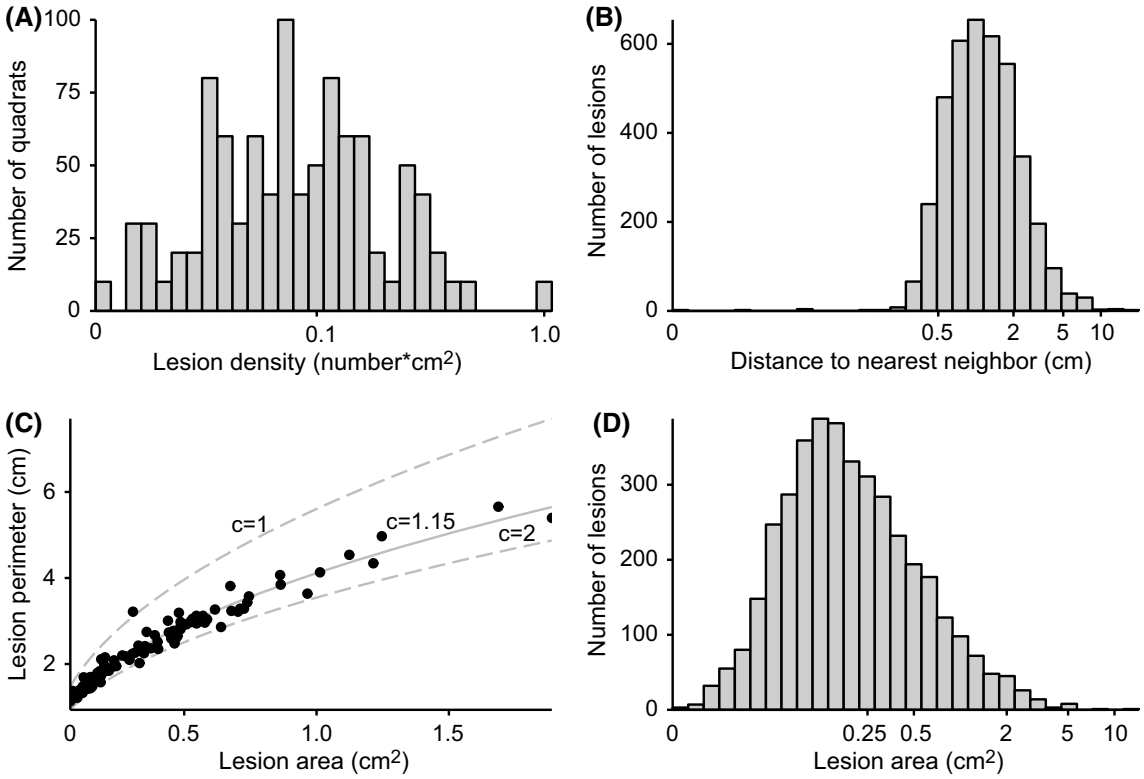

Fig. 1 Spatial patterning of coral lesions, giving **a** the density (number of lesions per live area of coral) among quadrats, **b** the distance to the centroid of the nearest neighboring lesion, **c** the shape of the lesions, and **d** the size of the lesions. In panel **c**, the dashed lines represent predictions for $c = 1$ (equal elliptical axes), and $c = 2$ (the major elliptical axis is twice that of the minor). Results are based on 97 quadrats (**a**, **c**), and 3957 lesions (**b**, **d**). All histogram x axes are on the log10 scale

Table 1 Model selection information—field surveys of coral lesions

Model	Distribution	Log likelihood	AIC	Δ AIC
Coral area + site	Negative binomial	−427.02	864.04	0
Coral area	Negative binomial	−435.50	875.01	10.97
Coral area + site	Poisson	−1783.40	3576.80	2712.77
Coral area	Poisson	−2081.63	4167.26	3303.23

Models were fit for the number of coral lesions at the quadrat level. The best model included both the area of live tissue, site, and an overdispersion term indicating spatial clustering

experiment ($t_{35} = -0.63$, $P = 0.53$). Additionally, spacing between lesions did not significantly change colony growth at either time point.

Effect of acute damage distribution on coral linear extension

After 1 year, the linear extension of corals depended on both the position on the coral colony and the damage pattern of the coral. The model that best described the data included a quadratic and cubic term (Table 2) as the relationship between coral growth and the angle of growth decreased nonlinearly from the top of the coral. Corals without any lesions (control) grew 1.32 cm at the top of the coral (i.e., 0°). For control corals, linear extension decreased 19% for polyps that were 45° from the top of the colony, and by 54% at the point where the coral extended horizontally (i.e., at 90°, Fig. 4).

Corals with lesions far apart (centers separated by 3.5 cm) showed only a 3% reduction in growth compared to the undamaged corals, and this reduction across the colony was not significant (Table 3, Fig. 3). In contrast, corals that had two lesions in close proximity of one another (lesion centers separated by 1.2 cm) showed a 18% reduction in growth on the top of the coral compared to control corals, but no difference beyond 45° (Fig. 3). Thus, the response of a coral to damage after 1 year depended on the spacing between lesions—effects were only apparent in the region of the damage and only when lesions were in close proximity to one another (i.e., within 2 cm).

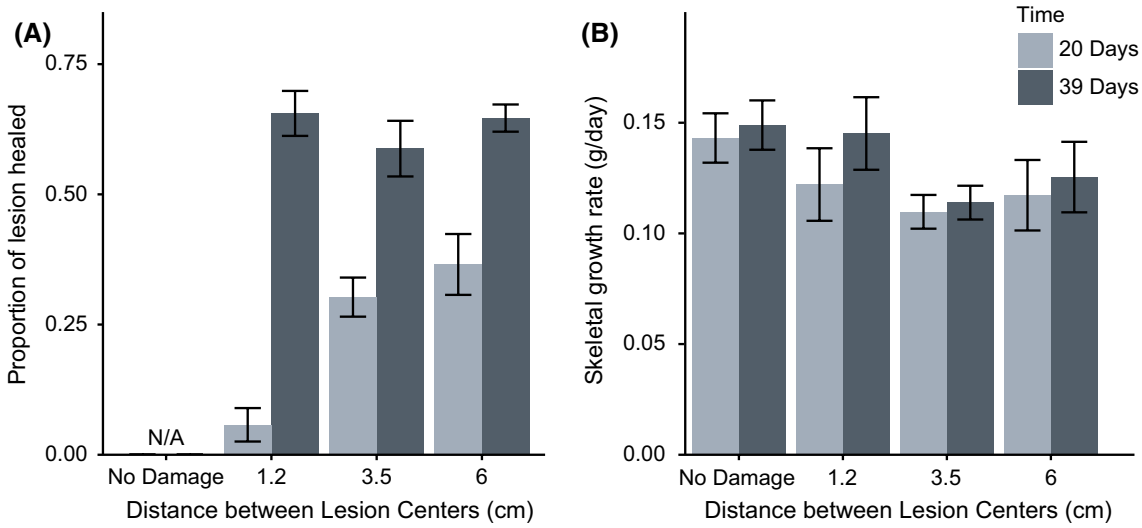

Fig. 2 Effect of lesion spacing on **a** percent of lesion healed and **b** coral skeletal growth rate and after 20 (light gray bars) and 39 (dark gray bars) days. The percent of lesion healed was significant between corals with lesions with centers separated by 1.2 cm and by at least 3.5 cm after 20 days, but not 39. There were no significant effects between distance between lesion centers and skeletal growth. Values plotted are fitted values from the mixed effects model and error bars represent ± 1 SE ($N = 10$)

Table 2 Model selection information—effect of lesion spacing on linear extension

Model	Log likelihood	AIC	Δ AIC
Treatment + angle^2 + angle^2 + treatment × angle^2 + treatment × angle^3	298.26	−574.52	0
Treatment + angle + treatment × angle	278.74	−541.49	33.04
Treatment + angle + angle^2 + angle^3 + treatment × angle + treatment × angle^2 + treatment × angle^3	283.22	−538.44	36.09
Treatment + angle + angle^2 + treatment × angle + treatment × angle^2	278.34	−534.68	39.84
Treatment + angle^3 + treatment × angle^3	264.33	−506.67	67.85
Treatment + angle^2 + treatment × angle^2	114.37	−212.73	361.79

All models include the individual coral as a random effect

Table 3 Parameter estimates—effect of lesion spacing on linear extension

Effect	Value	Std. error	*t* value	*p* value
Intercept	1.32	0.063	20.90	< 0.0001
Near	−0.24	0.10	−2.53	0.016
Far	−.04	0.091	−0.43	0.67
Angle^2	−0.00016	< 0.0001	−25.41	< 0.0001
Angle^3	< 0.0001	< 0.0001	15.28	< 0.0001
Near × angle^2	0.0001	< 0.0001	8.79	< 0.0001
Far × angle^2	−0.00001	< 0.0001	−1.62	0.11
Near × angle^3	≥ 0.0001	< 0.0001	−7.22	< 0.0001
Far × angle^3	< 0.0001	< 0.0001	2.81	0.0049

Effect of chronic damage on coral morphology

I observed differences in the change in coral morphology based on the presence of the corallivorous snail, *C. violacea*. Near the snail, there was an increase in the topographic complexity, while in areas away from the snails, topographic complexity decreased (t_{28}, $P = 0.047$, Fig. 4a). Linear extension of coral was reduced by 68% in the presence of *C. violacea* ($t_{18.5}$, $P = 0.0023$, Fig. 4b). These results support the hypothesis that snails increase the heterogeneity of coral morphology by decreasing linear extension rates of the coral near where they feed and that this effect is quite localized (e.g., much smaller than the scale over which I measured topographic heterogeneity, 20 cm).

Discussion

The spatial patterning of lesions (i.e., how close lesions are to neighboring lesions) plays an important role in the recovery of a coral from damage. Lesions near one another heal more slowly than those further apart (Fig. 2a) and over longer time periods (a year), the spatial patterning of the

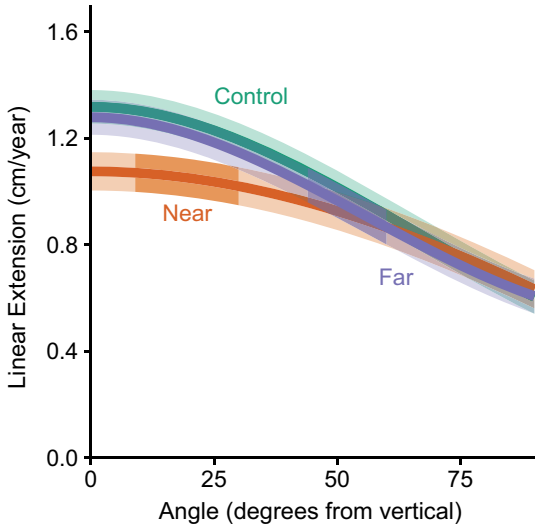

Fig. 3 Effect of lesion spacing [control: no lesions ($N=13$), near: lesion centers separated by 1.2 cm ($N=10$), far: 3.5 cm ($N=12$)] on coral growth rates at different locations on the colony (represented by the angle of growth). Lines give best fits, lightly shaded regions indicate ± 1 SE, and darkly shaded regions indicate the average lesion locations in each treatment (corals had two lesions, with one wound located to each side of 0°)

damage differentially affect skeletal growth patterns across the surface of the coral. When lesions are close together, there is a local reduction in linear extension near the lesion location, while corals with lesions far apart only show a slight decrease in linear extension compared to corals without damage (Fig. 3). Finally, chronic damage, such as that produced by the corallivorous gastropod, *C. violacea*, not only reduces linear extension of coral, but also has the potential to affect the shape of corals, creating mosaics of depressed growth (where snails feed) and comparatively elevated growth (away from the corallivore), resulting in more topographically complex corals (Fig. 4). Because lesions are generally very near one another (Fig. 1), these findings suggest that the effect of coral damage is more severe than effects obtained from studies of single lesions, resulting in slower healing rates and longer-term decrease in skeletal growth.

These results likely arise because resources are shared among polyps in a local neighborhood. The presence of nearby, healthy polyps facilitate the regeneration of tissue in areas that have been damaged (Bak 1983; Meesters et al. 1992). The ability of a polyp to contribute to the healing of a neighbor is likely affected by its proximity to the area of damage, as well as its own health, and the health of nearby polyps. These findings build on past studies that demonstrate this relationship between available resources and tissue regeneration. For example, wounds with a greater perimeter: surface area heal more quickly than similarly sized lesions (but with small perimeters) because there is more healthy tissue bordering the lesion that can extend in and heal the lesion (Meesters et al. 1997; Oren et al. 1997a). Similarly, smaller lesions heal more quickly than larger lesions (Henry and Hart 2005).

Previous studies have noted that lesions in clusters heal more slowly than single lesions and lesions in higher densities are more likely to grow in size than lesions in low densities (Welsh et al. 2015). Of course, these results could simply result from more damage created when lesions are

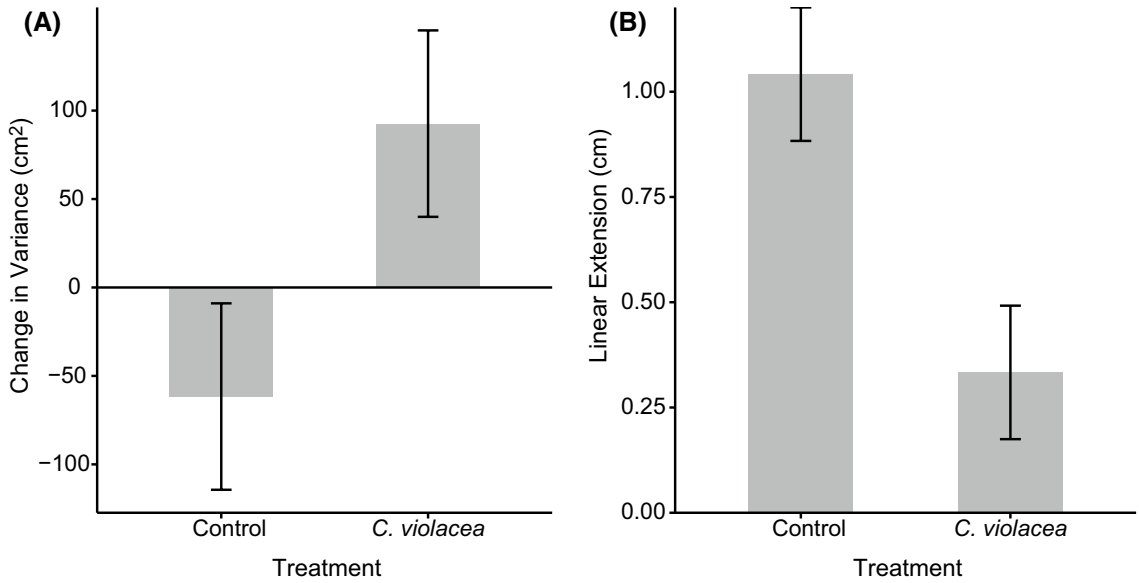

Fig. 4 Effect of damage on coral morphology. **a** *C. violacea* are significantly associated corals increasing in topographic complexity (higher variance), and facilitate crevices by significantly decreasing linear extension near their feeding site (**b**). Error bars represent ± 1 SE ($N=12$)

clustered, similarly to how corals with more gastropod corallivores present exhibit lower growth rates (Hamman 2018). Additionally, two adjacent lesions that effectively form one large lesion also exhibit lower regeneration rates (DeFilippo et al. 2016), but this study highlights how the spatial configuration of discrete lesions affects coral response. Future work should examine interactions between lesion characteristics (e.g., lesion size and shape) and spatial configuration.

The spatial patterns of damage likely interact with characteristics of the lesion due to their combined effects on the ability of polyps to share resources. For example, previous studies demonstrate the effect of size and shape on healing (e.g., Meesters et al. 1997; Oren et al. 1997a, b; van Woesik 1998), so it is likely that corals with elliptical lesions far apart from one another will likely regenerate tissue much more quickly than circular lesions near one another.

Other factors could affect the ability of units within a colonial organism to share resources. Corals in different environments can exhibit different regeneration times (e.g., Lester and Bak 1985; Sabine et al. 2015) and stressors such as sedimentation (Sheridan et al. 2014) and disease (Ruiz-Diaz et al. 2016), can reduce the overall health of the colony and therefore decrease the resources that neighbors have available to heal damaged units, and would therefore decrease the ability of the colony to recover from a damage event. For example, studies tracking isotopes demonstrated that bleached corals have lower connectivity [i.e., fewer isotopes were transported across the colony (Fine et al. 2002)], and lower rates of lesion healing, compared to unbleached colonies (Meesters and Bak 1993; Fine et al. 2002). If fewer resources are available due to stressors other than corallivory, the effects of configuration might differ given the degree of integration within the colony.

Lesion regeneration differs among coral species (Bak and Steward-Van Es 1980; Hall 1997), including differences between perforate corals share resources much more readily than imperforate corals (Cameron and Edmunds 2014), so the effect of lesion configuration will likely differ based on coral species. An imperforate coral such as *Pocillopora* spp. will likely not demonstrate the same effects as massive *Porites* due to limited resource sharing. Therefore, these effects are likely more important for corals that share resources over a larger range.

While I failed to detect an effect of spacing on short-term coral calcification (see Fig. 2), other studies have detected a trade-off between skeletal growth and tissue regeneration [see review in (Henry and Hart 2005)] suggesting that reallocating resources from calcification to tissue regeneration reduces the coral's ability to calcify for a certain time period following regeneration. However, other studies demonstrate compensatory growth in corals following damage (e.g., Jayewardene 2010). I observed neither, although there was a trend towards a trade-off (Fig. 2b). Other work notes effects of corallivory on defensive characteristics such as nematocyst densities (Rotjan and Dimond 2010), as well as reproduction (Van Veghel and Bak 1994) that might affect energetic tradeoffs and long-term patterns.

I detected a decrease in linear extension near lesions after 1 year for lesions located near one another (Fig. 3), but this effect was localized to the area near the lesions, whereas corals with lesions far from one another showed no significant difference from corals without any lesions. This suggests that longer-term effects of damage might be limited to certain spatial distributions of damage due to limitations of the coral to transport resources. Many lesions are created by consumers that target certain areas of the coral. For example, some fish frequently feed on tops of colonies (Rotjan and Lewis 2008; Welsh et al. 2015) and smaller organisms, such as the gastropod *Drupella cornus* juveniles seek out crevices for protection from predators (Schoepf et al. 2010). Other types of organisms do not feed on coral, but inflict other types of damage. For example, macro-algae abrade coral (Jompa and McCook 2003), and vermetid gastropods deploy nets that contact coral and reduce growth (Shima et al. 2010, 2015). In these cases, damage is more likely to occur on more "peaked" parts of the coral, where contact with foliose algae or vermetid nets is more likely.

Depending on the location of damage, these organisms could facilitate either topographically diverse corals (e.g., as I observed for *C. violacea*: Fig. 4) by decreasing growth in crevices, or could create less topographically complex corals by decreasing growth on the peaks (Shima et al. 2015). Given that coral species and morphology affect regeneration rates (Hall 1997), an interesting interaction might occur if more topographically complex corals (or "bumpier" morphotypes) are better suited to regeneration and skeletal growth following injury (Van Veghel and Bak 1994). These effects could not only alter coral growth and morphology, but the effects could feed back to create interesting dynamics given some organisms preferentially feed on certain parts of corals [e.g., parrotfish that feed preferential based on growth form and skeletal hardness (Littler et al. 1989)], and those that exploit coral morphology to shelter from predators (Schoepf et al. 2010).

While this study focused on the effects of damage on corals, other modular organisms are likely to be affected by the spatial distribution of damage. For example, many plants are modular, and function as a body of semi-autogenous agents (Oborny 2019), and overall growth is decreased in the presence of herbivory (Nykänen and Koricheva 2004), similarly to the observation in this study (Fig. 2b). For these modular organisms, nearby healthy ramets can aid ramets stressed or damaged by either environmental conditions (Wei et al. 2019) or herbivores (Chen et al. 2011). Therefore, the spacing of lesions and damage will likely play a role in the recovery of these plants as demonstrated in this study for corals.

While few studies have explicitly looked at the role of damage spatial distribution in plants, inoculations of a fungus are more harmful further from carbon sources (Goodsman et al. 2013). Additionally, defoliation of *Pinus sylvestris* that was evenly spread (or only at the basal part of the shoot) did not affect new leaves, but defoliation only at the apical portion did (Honkanen et al. 1999). These results indicate that proximity to healthy parts of the plant is important for healing, as demonstrated in this study for corals as scars far apart heal more quickly than those close together (Fig. 2a).

Many other modular organisms also interact with consumers, just as corals interact with corallivores, perhaps in different ways than the coral system observed here. For example, while many herbivores may not be attracted to plant damage itself (Nykänen and Koricheva 2004), others either are attracted to or avoid pathogens (reviewed in Tack and Dicke 2013). The lack of attraction to plant damage suggests feedbacks among modular plants and their herbivores might be less likely to align with those in this study [massive *Porites* and *C. violacea*, where the consumer is attracted to damage, (Hamman 2018)]. However, when plants show induced defense, interactions might align with the effect of vermetid gastropods on coral morphology (Shima et al. 2015), where areas further from damage are more likely to be damaged than areas previously damaged. In fact, the movement of defensive chemicals to ramets of higher risk (e.g., younger ramets) in response to herbivory can lead to a modification in herbivory patterns as the herbivore's preference is countered (Chen et al. 2011). Given that herbivores have the potential alter plant traits [e.g., number of leaves and nitrogen content, (Ando and Ohgushi 2008)], spatial feedbacks between grazing and the response of the plant could produce interesting patterns in plant morphology and nutrient content.

In many ways, the patterns observed in this study, emerging from interactions among coral polyps, or potentially similar patterns of damage within a plant, mirror patterns that might be observed at much larger spatial scales, such as in meta-populations (White 1979). In meta-populations, extinct patches are colonized by occupied patches, similarly to how coral tissue is regenerated over a lesion via resources from nearby, healthy tissue. In a meta-population, colonists are the resource, and similarly to how the scale of coral regeneration depends on the scale of connectedness within the colony, meta-population persistence depends on the spatial scale of modularity (Fletcher Jr et al. 2013). Therefore, a single patch's ability to persist is driven by its connections to source patches. If damage or disturbance is clustered, then many extinct patches might be depending on colonists from the same patches leading to overall population extinction. In fact, theoretical studies do show overall population extinction when disturbance is spatially autocorrelated (Johst and Drechsler 2003; Kallimanis et al. 2005), indicating this study's empirical findings of the effects of damage clustering applies to not only other organisms, but potentially to other spatial scales and levels of organization.

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Compliance with ethical standards

Conflict of interest The author declares that there is no conflict of interest.

Ethical approval All applicable institutional and/or national guidelines for the care and use of animals were followed.

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